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Optimized Complex Convolutional Neural Network for Steady-State Visual Evoked Potentials Signal classification

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***Abstract*—The steady-state visual evoked potential (SSVEP) is one of the most commonly used paradigms in the BCI system, which produces a significant response to visual stimuli of a specific frequency. How to better improve the classification performance of SSVEP signals is currently the main task. In this paper, a modified C-CNN-SVM model of EEG signal is proposed, aiming to enhance the precision of SSVEP signal classification. Firstly, spectral features are obtained from the SSVEP signal using fast fourier transform (FFT) as input to the model. FFT obtains 128 data lengths of the SSVEP signal, including 64 real part data and 64 complex part data. Then, CNN is used to further extract features, and finally, support vector machine (SVM) is used for classification. The experimental results show that the modified C-CNN-SVM model achieves an average accuracy of 93.86%, which is better than the traditional CCA, FBCCA and TRCA algorithms.**

Keywords—bci, ssvep, cnn, c-cnn, svm

I. INTRODUCTION

Brain-computer interfaces (BCIs) technology is a technology that translates brain activity into commands to control external devices[1]. In the EEG(Electro encephalo gram) signals caused by visual stimuli, The fundamental and harmonic components are the key basis for achieving SSVEP signal classification. Exploring the classification methods of SSVEP is crucial for the growth of the field of BCIs, so analyzing the accuracy of SSVEP is of great significance.

Canonical Correlation Analysis (CCA) is a multivariate statistical method that is effective in identifying complex correlation patterns between variables in two datasets. In recent years, owing to the benefits of efficiency, resilience, and ease of application, CCA algorithm has been widely used in SSVEP signal analysis[2]. In the BCIs system, Chen et al. studied SSVEP signals and used calibrated data for CCA detection, which can reduce misclassification caused by spontaneous SSVEP signals[3]. They proposed a Filter Bank -CCA method. Recently, Nakanishi et al. They have introduced a method of Task-Related Component Analysis (TRCA)[4]. Compared with CCA, TRCA uses SSVEP signals as templates, while CCA constructs templates using sine and cosine functions.

There are currently many methods for processing SSVEP signals, all of which use neural networks to analyze SSVEP signals. Convolutional neural networks (CNNs) which performs excellently in image and video recognition, classification, and feature extraction. Due to its ability to automatically extract features from images, this network has independence in feature extraction and improves its ability to capture image details, CNNs have the significant advantage of processing EEG data. Afterwards, Roy Y et al. proposed a deep learning method called DCNN for analyzing SSVEP signals from 40 spelling task stimuli[5], the accuracy reaches 87%.

By analyzing the related literature, it is found that SSVEP classification has been studied with CNNs, and these studies used magnitude spectrum features as inputs to CNNs. However, this amplitude spectrum based method cannot fully utilize the phase information in SSVEP EEG signals, which limits the performance of CNN. To overcome this limitation, in this paper, we use FFT to obtain spectral characteristics, Then use CNN combined with support vector machines to classification SSVEP signals. The results verified the effectiveness of the proposed methods.

II. DATASET

A. Experimental Flow

The SSVEP signals we used was collected in the laboratory. T The user interface adopts a display screen with a 60Hz refresh rate, with a screen ratio set to 16:9. One of the nine white stimulus blocks on the display will randomly appear as a red stimulus block, and the subject will need to prepare for the stimulus task that will begin next. Nine flashing stimulus blocks on the monitor were simultaneously flashed at the corresponding fixed frequencies and phases (frequencies: 9.25 -14.25 Hz, phases: 0-1 π) continuously for 4 s. To reduce electrical signal interference caused by eye movements[6], the subjects were required to always gaze at the previously cued red stimulus block.

The experiment was divided into 15 blocks, and each block contained nine experimental trials that evoked the SSVEP signals. Each trial was divided into three steps in chronological order: 0s~1s for rest; 1s~2s for prompt, and 2s~4s for stimulation, as shown in Fig 1.

We deployed eight electrodes in the parietal and occipital regions(O1, O2, Oz, POz , PO3, PO4, PO5, PO6) of the brain for SSVEP signal acquisition, as shown in Fig 2.

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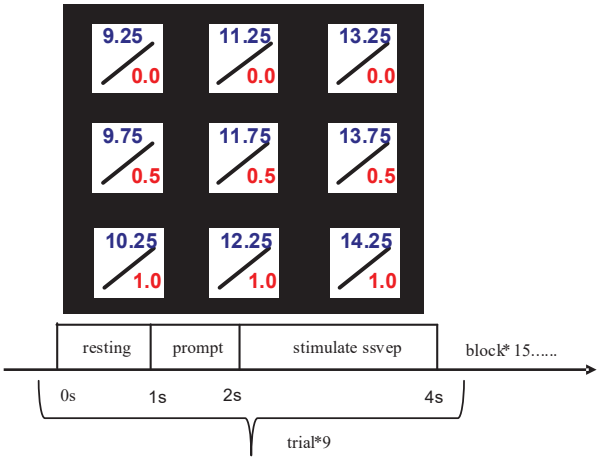


Fig. 1. Flow chart of the experiment

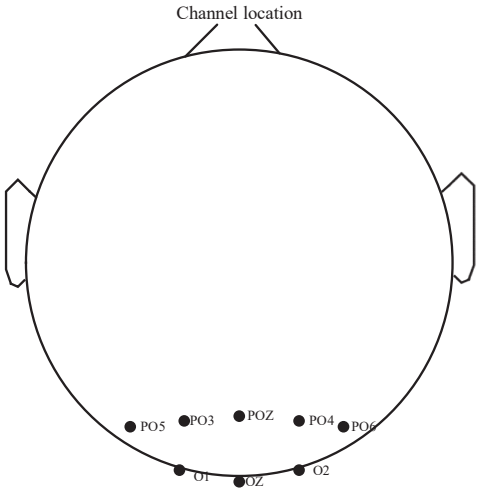


Fig. 2. Electrode placements according to the International 10-20 system.

B. Preprocessing

The data were collected by the Neuroscan 64 channels EEG acquisition system. All subjects were normal visual acuity and corrected visual acuity levels, physically healthy, no psychiatric disorders, no cognitive deficits, non-smokers, non-drinkers of alcohol and coffee, and normal hearing and articulation skills. 10 subjects (5 males,5 females, average age: 23.7 ± 0.61 years old) had their SSVEP signal collected, at a sampling rate of 1000 Hz, and the collected SSVEP signals need to be pre-processed, including down sampling, deleting useless electrodes, and eliminating noise, etc. In this paper, we filtered the SSVEP signals with a band-pass filter of 3-35 Hz, and down sampled them to 250 Hz. The experiments were conducted to analyze the SSVEP signal data of 2s[7], and the data was divided into a training set. The SSVEP signal data of 2s is analyzed. The data were divided into a training set, a validation set and a test set, then the proportions of dividing the dataset are 60%, 20% and 20% of the total SSVEP dataset.

III. METHODOLOGY

A. Complex Spectrum Features

In C-CNN, the standard FFT is first used to set the resolution, which is 0.5 Hz based on the characteristics of the SSVEP signal. The input time-domain signal is then converted into a complex FFT representation. Then extract the frequency components of the real and imaginary parts from 3 Hz to 35 Hz between each channel, and obtain two data lengths with a value of 64. Connect these two lengths of data into a feature vector, where the first half contains the real part of the FFT and the second half contains the imaginary part of the complex FFT. Then, the obtained feature is superimposed under another feature to form a matrix of $N_{ch} * N_{fc}$, where N_{ch} is the channel number and N_{ch} is the value of the frequency domain signal, which is equal to 128. The CNN method used by taking the complex results of the FFT as input is called the C-CNN method, as shown in Equation (1).

$$I_{C-CNN} = \begin{bmatrix} Re\{FFT(x_{O1})\}, Im\{FFT(x_{O1})\} \\ Re\{FFT(x_{O2})\}, Im\{FFT(x_{O2})\} \\ Re\{FFT(x_{Oz})\}, Im\{FFT(x_{Oz})\} \\ Re\{FFT(x_{POz})\}, Im\{FFT(x_{POz})\} \\ Re\{FFT(x_{PO3})\}, Im\{FFT(x_{PO3})\} \\ Re\{FFT(x_{PO4})\}, Im\{FFT(x_{PO4})\} \\ Re\{FFT(x_{PO5})\}, Im\{FFT(x_{PO5})\} \\ Re\{FFT(x_{PO6})\}, Im\{FFT(x_{PO6})\} \end{bmatrix} \quad (1)$$

B. Convolutional Neural Network

We obtain the complex spectral features of the SSVEP signal through FFT, and then input the features into the CNN. Using two convolutional layers of CNN: the first layer serves as a spatial filter layer to capture spatial features between different electrodes; The second layer serves as a time filter layer to extract the temporal dynamic information of the signal. Through this approach, CNN can effectively learn feature patterns in the data, and finally, these features are used in SVM for signal classification. The CNN architecture is shown in Fig. 3.

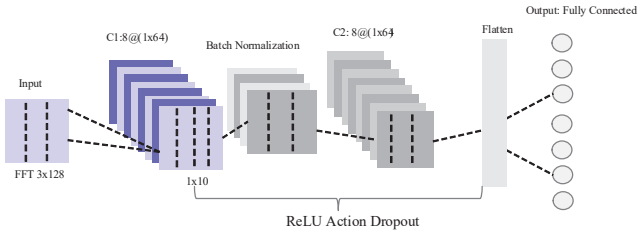


Fig. 3. CNN model architecture

Deep learning models learn features and classifiers from data through training, and deep learning methods offer the advantage of automatic feature extraction over classical algorithms that traditionally use a feature extraction step as input to the classifier (i.e., convolutional neural networks for classification tasks). CNNs using frequency-domain inputs have a similar but relatively simple network structure and reduced computational complexity compared to CNNs using time-domain inputs.

C. Support Vector Machine

Support vector machine commonly used for classification and regression tasks. Its main task is to find the optimal hyperplane and then partition different types of data points to maximize the distance between the data points and the hyperplane. For example, we have a dataset[8].

$$\{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$$

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x_i is the input feature vector, and in this paper we use y_i as the corresponding category label, for multi-class problems, y_i belongs to more classes and the SVM needs to be extended appropriately. The goal of a SVM is to find a hyperplane $w^T x_i + b = 0$, where w is the normal vector and b is the bias term, such that the interval between two parallel hyperplanes can be maximized and all training samples can be classified correctly[9]. This interval is the maximum boundar. The optimization problem for a support vector machine can be formalized as a convex quadratic programming problem as in Eqs. (2) and (3):

$$\min_{w,b} \frac{1}{2} \| w \|^2 \tag{2}$$

$$y_i (w^T x_i + b) \geq 1 \, for \; i = 1, 2, \ldots, n \tag{3}$$

$\| w \|$ is the paradigm of the weight vector, x_i is the ith sample in the training set, y_i is the corresponding category label. In multi class SVM, we usually adopt several strategies to extend the basic binary SVM, including one to many (One vs All, OVA), one to one (One vs One, OVO), or methods using multi class kernel functions. Each method has its own specific optimization problem and solution method This article uses a one to many approach, but the core goal is still to find a hyperplane that can maximize boundaries and correctly classify training samples[10]. The dual problem of multi class SVM is solved through Lagrange multipliers and KKT conditions, transforming the original problem into a dual problem, and then using optimization algorithms for solution.

In nonlinear multi-class problems, SVM maps raw data to a higher dimensional space through kernel functions, in order to find linear hyperplanes that can distinguish different categories in that space[11]. Although the direction and position of the hyperplane are invariant to the scaling of (w, b), the output value y of the decision function will change with the variation of (w, b). This means that for each category in the multi-class problem, we can find a specific hyperplane that maximizes the separation of samples from other categories. As shown in Fig 4. By adjust (w, b) appropriately, it can be ensured that the boundary points of each category satisfy specific decision function values, such as $\omega^T x + b = -1$ and $\omega^T x + b = 1$, which can be considered as decision boundaries for different categories[12]. In this way, different hyperplanes work together to divide the feature space into multiple regions, each corresponding to a specific category, thus achieving multi-class tasks.

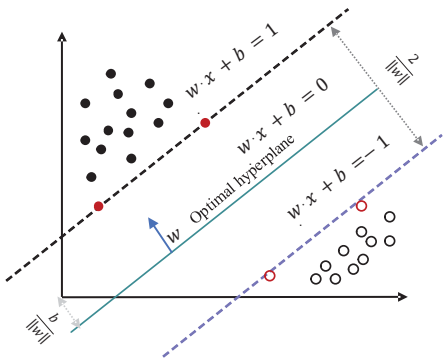


Fig. 4. SVM multi-class

D. C-CNN-SVM MODEL

Based on the existing C-CNN model, the modified C-CNN-SVM model is proposed, as shown in Fig. 5.

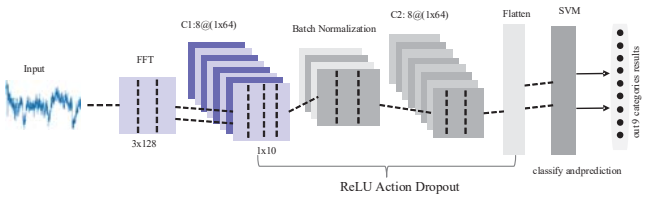


Fig. 5. Diagram of C-CNN-SVM framework

A model is composed of several levels, one of which is a simple linear stacked model that can add various neural network layers in order, as shown in Table 1. The first layer is the input layer, which inputs data into the model. A 2D convolutional layer with two filters, each with a size of N_{ch} *1. Batch normalization layer, batch normalization helps accelerate the training process and can improve the model's generalization ability. The activation function uses the ReLU activation function. Dropout layer, used to randomly discard some neurons during training to prevent overfitting Droprate is the drop rate. The second 2D convolutional layer, unlike the first convolutional layer, has a filter size of 1 *'$kernel_f$ ' This line adds a Flatten layer to convert two-dimensional feature maps into one-dimensional vectors. This transformation is done to interconnect the feature vectors with the SVM classifier in preparation for the classification task.

TABLE I. C-CNN-SVM PARAMETERS

Number of layers	Type of level	Filter	Kernel	Activation function
1	input layer			ReLU
2	2D Convolutional Layer	256	10	
3	2*CNN_PARAMS			
4	batch normalisation layer			
5	dropout layer		0.25	
6	2D Convolutional Layer	256	10	ReLU
7	Flatten layer			
8	SVM			

This article first uses C-CNN to extract features and denoise data, and uses the two convolutional layers of the model to learn spatial and temporal features, respectively. The output of the CNN is usually the feature vectors, which are used as inputs to the SVM. SVM distinguishes different categories by finding the best hyperplane in the feature space that maximizes the boundary between different categories. During the training process, the classification performance of SVM is optimized by adjusting the kernel function parameters and cross-validation of SVM. SVM uses the trained feature vectors to differentiate between different data categories, thus achieving accurate classification of SSVEP signals.

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IV. Algorithm Validation Experiment and Result Analysis

In this paper, we used PyCharm as the development environment and constructed a C-CNN-SVM model using the Keras deep learning library to classify SSVEP signals. We have adjusted the model parameters to optimize feature extraction and classification performance. To evaluate the generalization ability of the model, we adoption 10x cross validation and selected the cross-entropy loss function and SGD optimizer to guide model training. In addition, we adjusted the kernel function of support sector machine to enhance the classification accuracy of the model in high-dimensional feature space.

Our results show that the C-CNN-SVM model is generally superior to the C-CNN model, with an average accuracy of 93.86%, and is superior to traditional algorithms such as CCA, FBCCA, and TRCA, as shown in Table 2. All results are derived from our collected data.

TABLE II. SSVEP Signals Classification Accuracy

model / subject	CCA	TRCA	FBCCA	C-CNN	C-CNN-SVM
S1	22.22	41.67	72.5	75.69	93.13
S2	22.78	56.94	50.83	76.80	90.12
S3	32.22	78.47	86.67	93.43	94.57
S4	76.11	97.92	100	98.61	96.35
S5	68.33	99.31	99.17	99.72	93.13
S6	76.11	95.83	95	99.72	95.13
S7	62.22	86.81	99.17	92.63	91.69
S8	91.11	93.06	100	99.02	97.24
S9	53.89	96.53	97.5	97.36	95.46
S10	56.44	79.86	84.17	90.27	91.69
Mean±Std	56.44± 22.47	82.64± 18.39	88.51±15 .21	92.33± 8.65	93.86±2. 21

Fig 6 shows, we used confusion matrix analysis on the SSVEP signals of the subjects to verify the feasibility of the improved combination model.

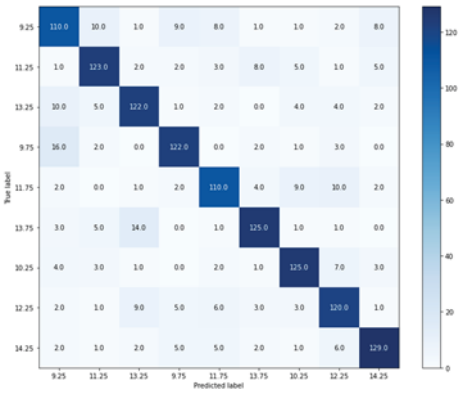


Fig. 6. C-CNN-SVM confusion matrix

V. Conclusions

In order to improve the classification of analyzing SSVEP signals, this paper proposes a combined model for EEG signal analysis based on C-CNN-SVM. This combined model utilizes the amplitude and phase information of frequency domain signals, combined with CNN and support vector machine, to classification SSVEP signals. By comparison of different algorithms, it is found that the C-CNN-SVM model achieves an average accuracy of 93.86%, which is better than C-CNN, the traditional CCA, FBCCA and TRCA algorithms. The combination of CNN and SVM, which is of great significance for analyzing SSVEP signals. In addition, this method provides a new idea for SSVEP signal classification, which helps to promote the development and application of BCIs technology.

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